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OPERATING SYSTEMS

Concept Mapping

History and Major Milestones in Operating System Development

Operating system act as an interface between users, applications and hardware, managing resources efficiently and providing services for programs

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Operating System Development Evolution and Major Milestones

1. The Beginning
(1940s - early 1950s)

NO OS

- * Computers were huge
- * No operating system
- * Interacted directly with hardware

Key Milestones

- * manual operations
- * One job at a time

2. Batch Processing System (1950-1960s)

- * Jobs are collected and executed in batches
- * OS introduced to automate job sequencing and I/O operations

Key milestones

- * Batch processing
- * Better CPU utilization

3. Multiprocessing and Time Sharing (1960s - 1970s)

- * Multi programs kept in memory
- * Time-sharing allowed many users to interact with the system simultaneously

Key milestones

- * Multiproces users
- * Multi-tasking
- * UNIX

4. Personal Computing Era (1980s)

- * Rise of microcomputers and personal devices
- * OS became user-friendly with GUI and application support

Key milestones

- * GUI
- * Windows
- * Mac OS
- * MS-DOS

5. Networked and distributed systems (1990s)

- * OS evolved for networking, distributed computing and client-server environments
- * Focus on security, reliability and scalability

Key milestones

- * Linux - Open Source OS
- * Windows NT
- * Java OS

6. Modern Era - mobile cloud & virtualization (2000s - 2010s)

- * OS optimized for mobile devices, cloud platforms.
- * Emphasis on performance, energy efficiency and user experience

Key milestones

- * MAC OS with Aqua GUI
- * Android OS - mobile OS revolution

7. Contemporary & future Trends (2020s and Beyond)

- * OS supporting AI, IoT, edge computing
- * Greater integration, automation and intelligent resource management

Key milestones

- * AI integrated OS features.
- * Growth of edge OS for IoT devices